# INTERNAL BORDER GATEWAY PROTOCOL

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**Lab Introduction**

The purpose of this lab is to familiarize us with the configuration of IBGP over an internal routing protocol such as OSPF. We want to allow the routers to establish IBGP adjacency using their loopback addresses. The two routers outside of the IBGP area should be configured with external BGP. To complete this lab, we need to ping every loopback address along with showing IBGP adjacency between the routers in the IBGP area.

**Background information**

IBGP (Internal Border Gate Protocol) is a modification of an existing protocol called BGP (Border Gate Protocol). The purpose of BGP was to provide an extremely large scalable external gateway protocol that very large networks such as the internet will run on. BGP is meant to be extremely scalable where a router can store hundreds of thousands of prefixes without being extremely demanding on their processing power. Where IBGP comes in is in situations where it is critical to ensure routers are routing traffic reliably, which is where the full mesh topology of IBGP networks comes in (Where full mesh means every router has adjacency with every other router).

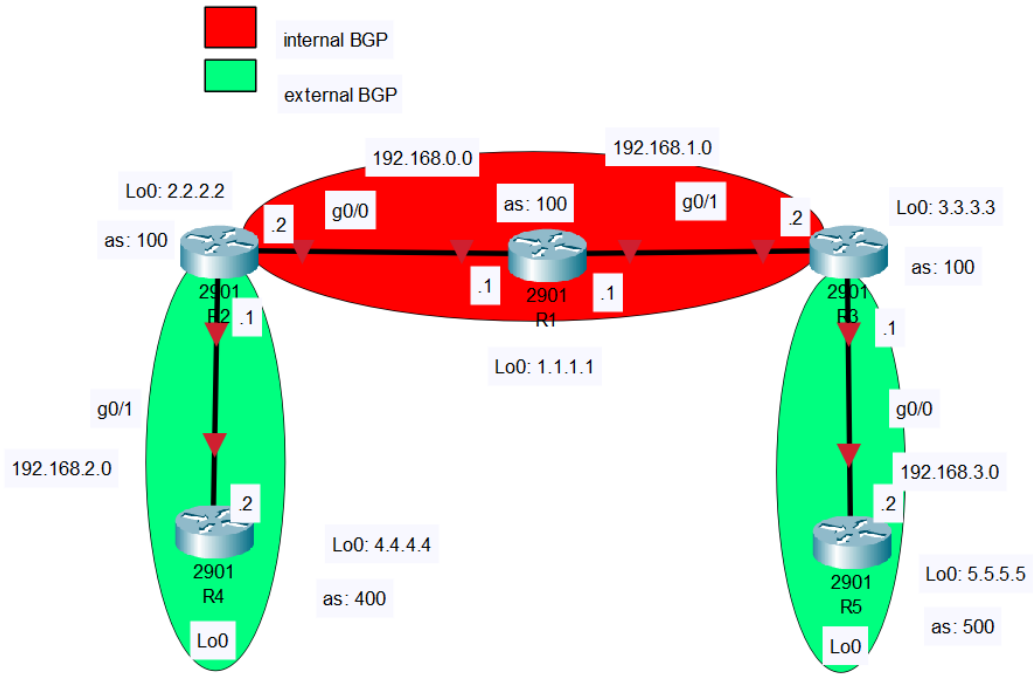
An example of this is a company with multiple ISPs. If the routers were running EBGP which do not use a full mesh topology (EBGP routers cannot establish adjacency with those not directly connected), if an ISP were to go down, routers directly connected to the ISP will only be able to forward their traffic towards routers that are directly connected. But <u>since this router does not actually have a route towards a different ISP network to reach the internet, the packets are instead dropped.</u> With IBGP, every router is fully meshed, which means that routes are updated by using IBGP. Thus, if an ISP were to go down, the routers connected to an end user will know if the ISP goes down because of the fully meshed topology constantly sending routing topology updates, and instead forward the traffic to a different ISP. Thus, the purpose of IBGP is to maintain redundancy and consistent end to end connectivity in one Autonomous system.

However, since the loopback addresses of IBGP routers are needed to establish adjacency, typically an IGP such as OSPF is used to provide routes towards the loopbacks to allow adjacency. The reason the full mesh is needed in IBGP is to prevent routing loops because IBGP has no loop prevention methods. Thus, large networks that are too impractical to make fully meshed will use route reflectors instead, where every route learned by an IBGP peer will immediately be sent to all other IBGP peers, negating the possibility of routing loops.

Unlike BGP, IBGP running routers can establish adjacencies with routers that are not directly connected. This is done through the update-source command, which specifies an

interface that is not directly connected. Loopback addresses are used instead of physical interfaces in BGP because if a physical interface goes down, BGP adjacency would disconnect. Loopback interfaces are not physical, and thus they can never go down which ensures that BGP adjacency will also never go down.

**Topology**




**Configuration**

## Router 1

```
hostname r1

interface Loopback0

ip address 1.1.1.1 255.255.255.255


interface GigabitEthernet0/0/0

ip address 192.168.0.1 255.255.255.0

negotiation auto
```

```
interface GigabitEthernet0/0/1
ip address 192.168.1.1 255.255.255.0
negotiation auto

interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto

interface Vlan1
no ip address
shutdown

router ospf 10
router-id 11.11.11.11
network 1.1.1.1 0.0.0.0 area 0
network 192.168.0.0 0.0.0.255 area 0
network 192.168.1.0 0.0.0.255 area 0

router bgp 100
bgp log-neighbor-changes
network 1.1.1.1 mask 255.255.255.255
network 192.168.0.0
network 192.168.1.0
neighbor 192.168.0.2 remote-as 100
neighbor 192.168.0.2 update-source Loopback0
```

```
neighbor 192.168.1.2 remote-as 100
neighbor 192.168.1.2 update-source Loopback0
```

## Router 2

```
hostname R2
interface Loopback0
ip address 2.2.2.2 255.255.255.255

interface GigabitEthernet0/0/0
ip address 192.168.0.2 255.255.255.0
no shut
negotiation auto

interface GigabitEthernet0/0/1
ip address 192.168.2.1 255.255.255.0
no shut
negotiation auto
ba 0
network 192.168.2.0 0.0.0.255 area 0

router bgp 100
bgp log-neighbor-changes
network 2.2.2.2 mask 255.255.255.255
network 192.168.0.0
network 192.168.2.0
neighbor 192.168.0.1 remote-as 100
neighbor 192.168.0.1 update-source Loopback0
neighbor 192.168.2.2 remote-as 400
```

```
neighbor 192.168.1.2 remote-as 100
neighbor 192.168.1.2 update-source lo0
```

## Router 3

```
hostname R3
interface Loopback0
ip address 3.3.3.3 255.255.255.255

interface GigabitEthernet0/0/0
ip address 192.168.3.1 255.255.255.0
no shut
negotiation auto

interface GigabitEthernet0/0/1
ip address 192.168.1.2 255.255.255.0
no shut
negotiation auto

router ospf 10
router-id 33.33.33.33
network 3.3.3.3 0.0.0.0 area 0
network 192.168.1.0 0.0.0.255 area 0
network 192.168.2.0 0.0.0.255 area 0
network 192.168.3.0 0.0.0.255 area 0

router bgp 100
bgp log-neighbor-changes
network 3.3.3.3
```

```
network 192.168.0.0
network 192.168.1.0
network 192.168.3.0
neighbor 192.168.1.1 remote-as 100
neighbor 192.168.1.1 update-source Loopback0
neighbor 192.168.3.2 remote-as 500
```

## Router 4

```
interface Loopback0
ip address 4.4.4.4 255.255.255.0

interface GigabitEthernet0/0/0
no ip address
negotiation auto

interface GigabitEthernet0/0/1
ip address 192.168.2.2 255.255.255.0
negotiation auto

router bgp 400
bgp router-id 44.44.44.44
bgp log-neighbor-changes
neighbor 192.168.2.1 remote-as 100

address-family ipv4
  network 4.4.4.4 mask 255.255.255.0
  network 192.168.2.0
  neighbor 192.168.2.1 activate
```

```
exit-address-family
```

## Router 5

```
interface Loopback0
ip address 5.5.5.5 255.255.255.0

interface GigabitEthernet0/0/0
ip address 192.168.3.2 255.255.255.0
negotiation auto

router bgp 500
bgp router-id 55.55.55.55
bgp log-neighbor-changes
neighbor 192.168.3.1 remote-as 100
address-family ipv4
  network 5.5.5.5 mask 255.255.255.0
  network 192.168.3.0
  neighbor 192.168.3.1 activate
exit-address-family
```

## Verification

### R1

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

      1.0.0.0/32 is subnetted, 1 subnets
C        1.1.1.1 is directly connected, Loopback0
      2.0.0.0/32 is subnetted, 1 subnets
O        2.2.2.2 [110/2] via 192.168.0.2, 00:05:53, GigabitEthernet0/0/0
      3.0.0.0/32 is subnetted, 1 subnets
O        3.3.3.3 [110/2] via 192.168.1.2, 00:10:24, GigabitEthernet0/0/1
      4.0.0.0/24 is subnetted, 1 subnets
B        4.4.4.0 [200/0] via 192.168.2.2, 00:01:41
      5.0.0.0/24 is subnetted, 1 subnets
B        5.5.5.0 [200/0] via 192.168.3.2, 00:02:59
      192.168.0.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.0.0/24 is directly connected, GigabitEthernet0/0/0
L        192.168.0.1/32 is directly connected, GigabitEthernet0/0/0
      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.1.0/24 is directly connected, GigabitEthernet0/0/1
L        192.168.1.1/32 is directly connected, GigabitEthernet0/0/1
O     192.168.2.0/24 [110/2] via 192.168.0.2, 00:01:43, GigabitEthernet0/0/0
O     192.168.3.0/24 [110/2] via 192.168.1.2, 00:10:24, GigabitEthernet0/0/1
```

```
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *>i 4.4.4.0/24       192.168.2.2              0    100      0 400 i
 *>i 5.5.5.0/24       192.168.3.2              0    100      0 500 i
 * i 192.168.0.0      192.168.0.2              0    100      0 i
 *>                   0.0.0.0                  0         32768 i
 * i 192.168.1.0      192.168.1.2              0    100      0 i
 *>                   0.0.0.0                  0         32768 i
 r>i 192.168.2.0      192.168.0.2              0    100      0 i
 r>i 192.168.3.0      192.168.1.2              0    100      0 i
```

R2

```
R2(config-router)#do show ip bgp
BGP table version is 10, local router ID is 2.2.2.2
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *>  4.4.4.0/24       192.168.2.2              0             0 400 i
 *>i 5.5.5.0/24       192.168.3.2              0    100      0 500 i
 *>  192.168.0.0      0.0.0.0                  0         32768 i
 * i                  192.168.0.1              0    100      0 i
 r i 192.168.1.0      192.168.1.2              0    100      0 i
 r>i                  192.168.0.1              0    100      0 i
 *   192.168.2.0      192.168.2.2              0             0 400 i
 *>                   0.0.0.0                  0         32768 i
 r>i 192.168.3.0      192.168.1.2              0    100      0 i
```

```
R2(config-router)#do show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

      1.0.0.0/32 is subnetted, 1 subnets
O        1.1.1.1 [110/2] via 192.168.0.1, 00:06:28, GigabitEthernet0/0/0
      2.0.0.0/32 is subnetted, 1 subnets
C        2.2.2.2 is directly connected, Loopback0
      3.0.0.0/32 is subnetted, 1 subnets
O        3.3.3.3 [110/3] via 192.168.0.1, 00:06:28, GigabitEthernet0/0/0
      4.0.0.0/24 is subnetted, 1 subnets
B        4.4.4.0 [20/0] via 192.168.2.2, 00:02:37
      5.0.0.0/24 is subnetted, 1 subnets
B        5.5.5.0 [200/0] via 192.168.3.2, 00:03:48
      192.168.0.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.0.0/24 is directly connected, GigabitEthernet0/0/0
L        192.168.0.2/32 is directly connected, GigabitEthernet0/0/0
O     192.168.1.0/24 [110/2] via 192.168.0.1, 00:06:28, GigabitEthernet0/0/0
      192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.2.0/24 is directly connected, GigabitEthernet0/0/1
L        192.168.2.1/32 is directly connected, GigabitEthernet0/0/1
O     192.168.3.0/24 [110/3] via 192.168.0.1, 00:06:28, GigabitEthernet0/0/0
```

R3

```
R3(config-router)#do show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

      1.0.0.0/32 is subnetted, 1 subnets
O        1.1.1.1 [110/2] via 192.168.1.1, 00:13:38, GigabitEthernet0/0/1
      2.0.0.0/32 is subnetted, 1 subnets
O        2.2.2.2 [110/3] via 192.168.1.1, 00:09:06, GigabitEthernet0/0/1
      3.0.0.0/32 is subnetted, 1 subnets
C        3.3.3.3 is directly connected, Loopback0
      4.0.0.0/24 is subnetted, 1 subnets
B        4.4.4.0 [200/0] via 192.168.2.2, 00:05:04
      5.0.0.0/24 is subnetted, 1 subnets
B        5.5.5.0 [20/0] via 192.168.3.2, 00:06:26
O     192.168.0.0/24 [110/2] via 192.168.1.1, 00:09:16, GigabitEthernet0/0/1
      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.1.0/24 is directly connected, GigabitEthernet0/0/1
L        192.168.1.2/32 is directly connected, GigabitEthernet0/0/1
O     192.168.2.0/24 [110/3] via 192.168.1.1, 00:05:06, GigabitEthernet0/0/1
      192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.3.0/24 is directly connected, GigabitEthernet0/0/0
L        192.168.3.1/32 is directly connected, GigabitEthernet0/0/0
```

```
R3(config-router)#do show ip bgp
BGP table version is 14, local router ID is 3.3.3.3
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *>i 4.4.4.0/24       192.168.2.2              0    100      0 400 i
 *>  5.5.5.0/24       192.168.3.2              0             0 500 i
 r i 192.168.0.0      192.168.0.2              0    100      0 i
 r>i                  192.168.1.1              0    100      0 i
 *>  192.168.1.0      0.0.0.0                  0         32768 i
 * i                  192.168.1.1              0    100      0 i
 r>i 192.168.2.0      192.168.0.2              0    100      0 i
 *   192.168.3.0      192.168.3.2              0             0 500 i
 *>                   0.0.0.0                  0         32768 i
```

R4

```
r4(config)#do show ip bgp
BGP table version is 7, local router ID is 44.44.44.44
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *>  4.4.4.0/24       0.0.0.0                  0         32768 i
 *>  5.5.5.0/24       192.168.2.1                            0 100 500 i
 *>  192.168.0.0      192.168.2.1              0             0 100 i
 *>  192.168.1.0      192.168.2.1                            0 100 i
 *   192.168.2.0      192.168.2.1              0             0 100 i
 *>                   0.0.0.0                  0         32768 i
 *>  192.168.3.0      192.168.2.1                            0 100 i
```

```
r4(config)#do show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

      4.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        4.4.4.0/24 is directly connected, Loopback0
L        4.4.4.4/32 is directly connected, Loopback0
      5.0.0.0/24 is subnetted, 1 subnets
B        5.5.5.0 [20/0] via 192.168.2.1, 00:05:45
B     192.168.0.0/24 [20/0] via 192.168.2.1, 00:05:45
B     192.168.1.0/24 [20/0] via 192.168.2.1, 00:05:45
      192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.2.0/24 is directly connected, GigabitEthernet0/0/1
L        192.168.2.2/32 is directly connected, GigabitEthernet0/0/1
B     192.168.3.0/24 [20/0] via 192.168.2.1, 00:05:45
```

R5

```
R5#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

      4.0.0.0/24 is subnetted, 1 subnets
B        4.4.4.0 [20/0] via 192.168.3.1, 00:05:54
      5.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        5.5.5.0/24 is directly connected, Loopback0
L        5.5.5.5/32 is directly connected, Loopback0
B     192.168.0.0/24 [20/0] via 192.168.3.1, 00:07:39
B     192.168.1.0/24 [20/0] via 192.168.3.1, 00:07:39
B     192.168.2.0/24 [20/0] via 192.168.3.1, 00:06:24
      192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.3.0/24 is directly connected, GigabitEthernet0/0/0
L        192.168.3.2/32 is directly connected, GigabitEthernet0/0/0
```

```
BGP table version is 7, local router ID is 55.55.55.55
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
              t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *>   4.4.4.0/24       192.168.3.1                            0 100 400 i
 *>   5.5.5.0/24       0.0.0.0                  0         32768 i
 *>   192.168.0.0      192.168.3.1                            0 100 i
 *>   192.168.1.0      192.168.3.1              0             0 100 i
 *>   192.168.2.0      192.168.3.1                            0 100 i
 *    192.168.3.0      192.168.3.1              0             0 100 i
 *>                    0.0.0.0                  0         32768 i
```

**Lab Commands**

neighbor (ip-address) update-source (loopback interface)

Attempts to establish IBGP adjacency with the neighbor IP address. Specifies the loopback interface to peer with.

**Problems**

IBGP Adjacency goes down when bgp process cleared

When running the command “clear ip bgp *”, IBGP process will turn idle indefinitely. We often had to completely re-configure bgp on the IBGP routers affected.

Update-source command doesn't work if you use it too quickly.

When running the update-source command, we had to wait until the neighbor's established adjacency before using update-source or IBGP can turn idle.

**Conclusion**

IBGP is very situational in use compared to its BGP variant. Whereas BGP is used to connect one autonomous system with hundreds of thousands of prefixes, IBGP is used to connect multiple autonomous systems together without the loss of prefixes. For example, redistributing BGP routes into OSPF. OSPF cannot hold as many routes as BGP because its scalability is very rigid, so we would have to redistribute BGP into IBGP which can run over OSPF to allow these routes to pass through. IBGP is appealing for situations where traffic control is paramount such as ISP networks. The full mesh topology of IBGP allows each router to know if any routes go down which is not possible for EBGP. The consistency of IBGP when connecting through loopback addresses also ensures that neighbor adjacency always is up. Thus, even if IBGP is not commonly used, it is still the best routing protocol to use when control over traffic is necessary for an autonomous system.

# IBGP & EBGP

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